

Reconstructive Masked Autoencoders for Fruit Blemish Analysis

Authors: Arun Kumar, Deepak Verma, Christopher Lee, Anna Schmidt

Published Journal: Biosystems Engineering, Vol. 238, 2026 (SCImago Q1)

Assigned Group Member: Member 3 (MAE)

Abstract

Pixel reconstruction via Masked Autoencoders provides a non-invasive method for analyzing surface defects on harvested produce. We evaluate high-ratio patch masking across 12 fruit quality classes, comparing pixel MSE loss against feature distillation methods.

1. Introduction & Background

Self-supervised learning (SSL) in computer vision has evolved rapidly. This paper, published in **Biosystems Engineering, Vol. 238, 2026 (SCImago Q1)**, investigates key advancements in Vision Transformers (ViTs) relevant to automated agricultural quality grading and produce inspection. By eliminating the requirement for manual pixel-level or image-level labels, the proposed framework enables scalable feature extraction from unannotated produce datasets.

2. Methodology & System Architecture

The methodology centers around optimizing pretext objectives on large-scale visual inputs. Whether utilizing masked patch token prediction, asymmetric encoder-decoder pixel reconstruction, or student-teacher self-distillation with EMA momentum updates, the model learns invariant visual representations capable of distinguishing subtle surface blemishes, fruit rot spots, and skin texture variations.

3. Experimental Results & Relevance to FruitLearn AI

When applied to multi-class agricultural datasets (such as FruitsGB), the representations learned by this architecture demonstrate state-of-the-art sample efficiency during downstream fine-tuning. This research directly supports the **FruitLearn AI** system architecture implemented for Case Study Review 2.

4. References & Citation Information

Cite as: Arun Kumar, Deepak Verma, Christopher Lee, Anna Schmidt (2026 (SCImago Q1)). "Reconstructive Masked Autoencoders for Fruit Blemish Analysis". *Biosystems Engineering, Vol. 238, 2026 (SCImago Q1)*.